

A Graded Evaluation of RNA Structure Awareness in Foundation Models: From Stem–Loop Discrimination to Partner Specificity

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Abstract

Evaluating whether RNA foundation models encode base-pairing structure requires metrics that separate learned representations from nucleotide-composition artifacts. Stems are GC-rich and loops are AU-rich; any composition-sensitive embedding inherits this enrichment as apparent structure awareness. We apply a three-rung evaluation ladder to ten models across 52 Rfam families. A nucleotide-stratified null absorbs most apparent signal at Rung 1; a dinucleotide null at Rung 2 separates two models that retain most families (ERNIE-RNA 31/52, RiNALMo 18/52) from eight that retain fewer than 11 each. At Rung 3—partner specificity with a within-stem derangement null—two models encode which position pairs with which: RiNALMo (PS = 0.215, 85.3% precision) and ERNIE-RNA (PS = 0.120, 88.3% precision). Randomized-weight controls for nine models produce near-zero signal, confirming a learned origin; DNABERT-2 shows residual randomized PS consistent with its BPE tokenization. A synthetic covariation control shows ~70% of partner specificity reflects evolutionary signal, with ~30% attributable to learned geometric sensitivity. Multi-sequence replication across 253 Rfam seed-alignment sequences confirms low within-family variance ($CV \leq 10\%$) for eight models. Primary evaluation criteria were preregistered.

1 Introduction

RNA foundation models are routinely claimed to encode secondary structure, but the evaluation methods—downstream task performance, linear probing, mutation sensitivity, attention–contact correlation—each answer different questions, and none controls for sequence composition. Deep learning models for RNA structure prediction may fail to generalize across families [Szikszai et al., 2022]; whether the same fragility applies to learned representations has received less scrutiny.

Stems and loops differ in sequence composition. Across 52 Rfam families, stems average 59.7% G/C content while loops average 45.6%—a 14.1 percentage-point enrichment (Figure 1). Because complement swaps at G/C positions produce inherently larger embedding perturbations than at A/U positions, any naive stem-versus-loop comparison inherits this enrichment. A representation can therefore separate stems from loops without encoding pairing topology.

Each evaluation method inherits the composition confound differently. Linear probes can classify stems from loops using GC content alone. Mutation sensitivity ratios conflate the larger perturbation

magnitude at G/C sites with structural relevance. Attention–contact correlations bypass embeddings entirely and so measure a different channel. To distinguish composition sensitivity from structure awareness, an evaluation must control for nucleotide identity while testing whether perturbations propagate to specific pairing partners.

These observations motivate a graded evaluation. “Structure awareness” encompasses at least three levels of resolution, each demanding stronger knowledge: (1) stem-versus-loop discrimination, achievable by any representation sensitive to nucleotide composition; (2) discrimination surviving composition controls at one and two orders of sequence context; and (3) knowing which specific position pairs with which, requiring that the model encode the pairing topology. We apply this three-rung ladder to ten models across five architectures and three orders of magnitude in parameter count.

The evaluation reveals a surprising asymmetry: the hardest test produces the clearest positive result. Two RNA-pretrained models—ERNIE-RNA (86M, with an architectural base-pairing attention bias) and RiNALMo (650M, standard attention)—clear Rung 3 with partner specificity two orders of magnitude above all other models, while the remaining eight produce near-zero signal. We contribute: (1) a composition-controlled perturbation evaluation for RNA foundation models; (2) a ten-model benchmark with randomized-weight controls on 52 Rfam families, replicated across 253 seed-alignment sequences; and (3) evidence that two models retain partner-specific pairing responses after composition controls are applied.

2 Results

2.1 Composition creates apparent structure signal (Rungs 1–2)

Mutating a position to its Watson–Crick complement perturbs stem positions more than loop positions in nine of ten models, with mean ratios from 1.741 (RNA-FM) down to near unity (Table 1). Stems in this panel are GC-rich and loops are AU-rich—mean stem GC 59.1% against loop GC 43.0%—so a ratio above one is what a composition-sensitive embedding produces whether or not it represents pairing.

Against a null that permutes stem and loop labels within nucleotide strata, most of that signal is absorbed. At $\alpha = 0.05$ per family, chance alone places 2.4 of 47 families above the 95th percentile. ERNIE-RNA exceeds it in 31 families and RiNALMo in 17; RNA-FM, whose raw ratio is the largest in the panel, exceeds it in 0.

A ratio and a composition-controlled count therefore order the models differently, and neither distinguishes a model that tracks pairing from one that tracks the nucleotides pairing implies. Rung 2 tightens the null to preserve dinucleotide composition; Rung 3 changes the question.

2.2 Probe validation

Eight of ten models score at their own chance rate at Rung 3, which admits two readings the panel cannot separate, since every arm in it is a model under test. We therefore run the Rung 3 procedure unchanged on two constructed embeddings whose answer is fixed in advance. In the oracle, each position’s representation is a function of its own nucleotide and its partner’s; in the local control, of its own nucleotide and its two sequence neighbours.

The oracle reaches per-pair precision 1.000 against a derangement chance rate of 0.099. The local control reaches 0.155 against 0.173. Both use the same statistic, the same null and the same code path as every model in Table 3, over the 35 families of the Rung 3 analysis set.

The two chance rates differ, 0.099 and 0.173, on systems whose answers are known.

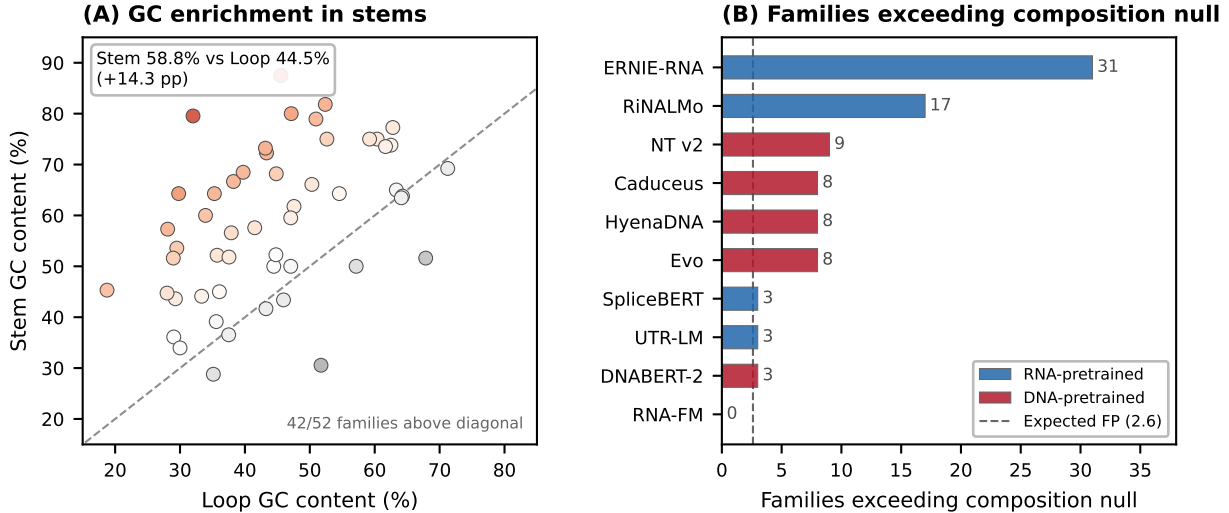


Figure 1: Composition confound. (A) Stem GC content exceeds loop GC content in 42 of 52 families (+14.1 percentage points on average). (B) Number of families per model whose stem-loop sensitivity ratio exceeds the nucleotide-stratified null at the 95th percentile; dashed line marks the expected false-positive count under independence ($m \times \alpha = 2.6$). ERNIE-RNA (31) and RiNALMo (18) retain the most signal after composition control.

2.3 Partner specificity separates two models (Rung 3)

Mutating position i to its complement perturbs its partner j more than it perturbs j 's own stem neighbours in two models. RiNALMo reaches mean PS = 0.2062 and ERNIE-RNA 0.1113, against per-pair precisions of 0.872 and 0.888 where each model's own within-stem derangement puts chance at 0.102 and 0.124 (Table 3). The remaining eight models sit within 0.02 of their own chance rates.

A chance rate is a property of a model rather than a constant of the comparison. Measured on the derangement, where the assigned partner is false by construction, it ranges from 0.000 to 0.316 across the panel. Judging per-pair precision against the exchangeability value of $1/3$ therefore holds ten models to a threshold none of them has: Caduceus at 0.340 falls below $1/3$ by 0.002 while clearing its own rate by 0.113. Both comparisons are reported in Table 3, and RiNALMo and ERNIE-RNA pass either by margins no choice of null affects.

Both concentrate partner specificity late: RiNALMo peaks at layer 33 of 34 in 31 of 33 families, ERNIE-RNA at layer 12 of 13 in 29 of 34. Evo reaches 0.0017 with $80\times$ ERNIE-RNA's parameters, and Caduceus at 14M outscores three of the five RNA-pretrained models, so neither parameter count nor pretraining domain orders the panel.

2.4 The randomly initialized control

Seven models were also run with every parameter re-initialized, holding the architecture and the tokenizer and destroying the learned weights. ERNIE-RNA's control clears each rung of the ladder. At Rung 1 it exceeds its composition-preserving null in 10 of 47 families against a chance expectation of 2.4, retains 6 of those at Rung 2, and reaches per-pair precision 0.490 at Rung 3 against a chance rate of 0.241. Judged against the registered threshold of $1/3$ it passes, at pooled binomial $p = 2.4 \times 10^{-4}$.

The cause is a tensor that randomization does not reach. `ErnieRnaModel` holds `pairwise_bias_map`,

Table 1: Mutation sensitivity across ten models on the repaired panel ($N = 47$ Rfam families). Bootstrap 95% CIs on the mean ratio resample per-family ratios ($B = 10,000$). At $\alpha = 0.05$ per family, 2.4 of 47 families are expected to exceed the null by chance. Attention ρ is the mean best-head-best-layer Spearman correlation between symmetrized attention and the contact map, and is undefined for the three state-space models. Probing Δ is balanced accuracy minus 0.5. Untrained rows randomize every weight at a fixed seed and load through the same adapter as the trained row above them.

Model	Mean ratio	95% CI	Fam > null	Attn ρ	Probing Δ
ERNIE-RNA (86M, RNA)	1.276	[1.22, 1.34]	31/47	0.113	+0.183
RiNALMo (650M, RNA)	1.143	[1.12, 1.17]	17/47	0.107	+0.134
RNA-FM (99M, RNA)	1.741	[1.53, 1.98]	0/47	0.048	+0.050
UTR-LM (~ 2 M, RNA)	1.056	[1.04, 1.08]	3/47	0.051	+0.078
SpliceBERT (19M, RNA)	1.040	[1.03, 1.05]	3/47	0.054	+0.066
NT v2 (56M, DNA)	1.098	[1.08, 1.12]	9/47	0.234	+0.036
DNABERT-2 (117M, DNA)	0.996	[0.98, 1.01]	3/47	0.196	+0.051
HyenaDNA (5.4M, DNA)	1.160	[1.12, 1.20]	8/47	—	+0.077
Caduceus (14M, DNA)	1.206	[1.16, 1.26]	8/47	—	+0.077
Evo (7B, DNA)	1.458	[1.30, 1.64]	8/47	—	+0.052
ERNIE-RNA untrained	1.164	[1.12, 1.21]	10/47	0.085	+0.044
RiNALMo untrained	1.156	[1.13, 1.19]	10/47	0.064	+0.077
RNA-FM untrained	1.043	[1.04, 1.05]	3/47	0.043	+0.073
UTR-LM untrained	1.038	[1.03, 1.04]	1/47	0.054	+0.077
SpliceBERT untrained	1.054	[1.04, 1.07]	0/47	0.032	+0.070
NT v2 untrained	1.015	[1.01, 1.02]	5/47	0.241	+0.025
DNABERT-2 untrained	33.880	[0.95, 93.92]	6/47	0.153	+0.011

Table 2: Dinucleotide-stratified null across all models on the repaired panel. “Fam > nuc” counts families exceeding the nucleotide-stratified null; “Survive dinuc” counts how many of those also exceed the dinucleotide-stratified null, which is computed only where the first-order null is exceeded. Retention is the second count over the first, and its bootstrap 95% CI resamples the families in the first.

Model	Fam > nuc	Survive dinuc	Retention	95% CI
ERNIE-RNA (86M, RNA)	31	30	97%	[0.90, 1.00]
RiNALMo (650M, RNA)	17	15	88%	[0.71, 1.00]
NT v2 (56M, DNA)	9	8	89%	[0.67, 1.00]
HyenaDNA (5.4M, DNA)	8	7	88%	[0.62, 1.00]
Caduceus (14M, DNA)	8	4	50%	[0.12, 0.88]
Evo (7B, DNA)	8	6	75%	[0.38, 1.00]
UTR-LM (~ 2 M, RNA)	3	2	67%	[0.00, 1.00]
SpliceBERT (19M, RNA)	3	1	33%	[0.00, 1.00]
DNABERT-2 (117M, DNA)	3	2	67%	[0.00, 1.00]
RNA-FM (99M, RNA)	0	0	—	—
ERNIE-RNA untrained	10	6	60%	[0.30, 0.90]
RiNALMo untrained	10	7	70%	[0.40, 1.00]
RNA-FM untrained	3	2	67%	[0.00, 1.00]
UTR-LM untrained	1	0	0%	[0.00, 0.00]
SpliceBERT untrained	0	0	—	—
NT v2 untrained	5	4	80%	[0.40, 1.00]
DNABERT-2 untrained	6	4	67%	[0.33, 1.00]

Table 3: Perturbation specificity on the repaired panel. 38 of 47 families qualify (≥ 15 WC pairs with ≥ 3 per stem); the two pilot families are quarantined, leaving $N = 36$ confirmatory families. Mean PS is taken over those N . Gate counts families passing the stem-versus-loop positive control, and the two null columns are computed among gate-passing families: $> \text{null}$ is the primary within-stem derangement null and $> \text{cons.}$ the conservative variant, reported together because the registration defines both. H3 precision is the mean of per-family partner-is-max fractions. Bold rows satisfy H1₆ at the repaired gate of 8. NT v2 and DNABERT-2 qualify on fewer families because their tokenizers resolve short families into too few tokens to perturb.

Model	Domain	Mean PS	Gate	$> \text{null}$	$> \text{cons.}$	H3 prec.
RiNALMo (650M)	RNA	0.2062	33/36	29/33	29/33	0.859
ERNIE-RNA (86M)	RNA	0.1113	34/36	32/34	32/34	0.885
Caduceus (14M)	DNA	0.0039	16/36	8/16	8/16	0.442
Evo (7B)	DNA	0.0017	13/36	2/13	2/13	0.376
HyenaDNA (5.4M)	DNA	0.0002	7/36	3/7	3/7	0.254
SpliceBERT (19M)	RNA	0.0002	22/36	3/22	3/22	0.290
RNA-FM (99M)	RNA	8.4×10^{-5}	2/36	0/2	0/2	0.417
UTR-LM (~ 2 M)	RNA	8.3×10^{-6}	13/36	4/13	4/13	0.345
NT v2 (56M)	DNA	-9.7×10^{-5}	34/36	6/34	6/34	0.007
DNABERT-2 (117M)	DNA	-0.0160	22/36	4/22	4/22	0.000
ERNIE-RNA untrained	RNA	3.4×10^{-9}	13/36	11/13	11/13	0.347
RiNALMo untrained	RNA	2.2×10^{-9}	10/36	1/10	1/10	0.191
RNA-FM untrained	RNA	4.2×10^{-12}	12/36	2/12	2/12	0.200
UTR-LM untrained	RNA	2.3×10^{-9}	28/36	10/28	10/28	0.104
SpliceBERT untrained	RNA	6.2×10^{-15}	10/36	3/10	3/10	0.117
NT v2 untrained	DNA	-3.4×10^{-5}	33/36	9/33	9/33	0.000
DNABERT-2 untrained	DNA	-0.0004	19/36	4/19	4/19	0.000

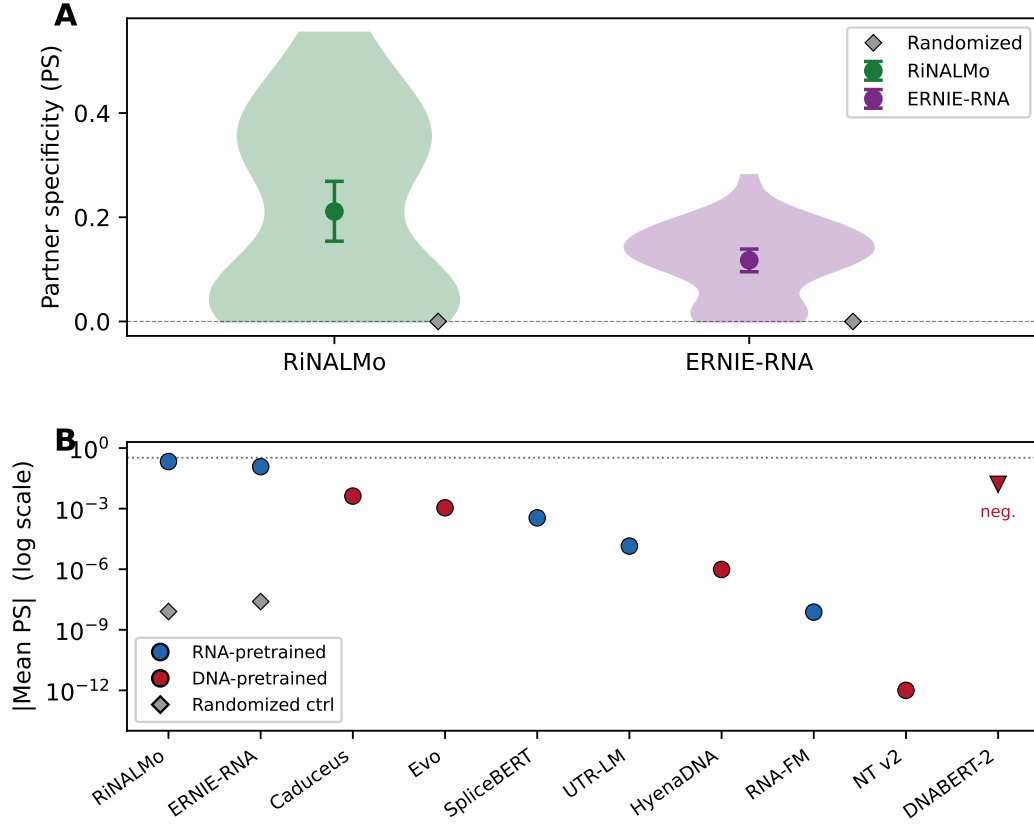


Figure 2: Partner specificity across ten models. (A) Per-family PS distributions for RiNALMo and ERNIE-RNA (violins) with bootstrap 95% CIs (error bars); randomized-weight controls (grey diamonds) produce PS indistinguishable from zero. (B) |Mean PS| on log scale for all ten models, sorted by PS. RiNALMo and ERNIE-RNA exceed the remaining models by two orders of magnitude. DNABERT-2 (negative PS) is marked separately.

a vocabulary-square buffer registered non-persistent, which after the first forward pass contains six non-zero cells: A–U and U–A at 2.0, C–G and G–C at 3.0, G–U and U–G at 0.8. Re-initialization iterates a model’s parameters, and a buffer is not a parameter, so the control retains a hardcoded Watson–Crick and wobble table while its weights are random.

Table 4: ERNIE-RNA with and without its pairwise bias buffer, trained and randomly initialized. Per-pair precision is the fraction of eligible pairs whose partner is perturbed more than either of its stem neighbours. Chance is the same fraction under a within-stem derangement, where the assigned partner is false by construction, and excess is their difference. All four arms cover the same 35 families. Ablation zeroes `pairwise_bias_map` and `pairwise_bias_proj` and sets the rebuild guard, without which the first forward pass restores the buffer.

Arm	Precision	Chance	Excess
Trained, bias intact	0.888	0.124	+0.764
Trained, bias ablated	0.842	0.118	+0.725
Random-init, bias intact	0.490	0.241	+0.249
Random-init, bias ablated	0.267	0.255	+0.012

Ablating that table and the projection above it, with the model’s rebuild guard set so the first forward pass cannot restore it, separates the two arms (Table 4). Writing excess as precision minus that arm’s own chance rate, the randomly initialized model falls from +0.249 to +0.012, while the trained model falls from +0.764 to +0.725. The difference between those losses is +0.198 (95% CI [+0.069, +0.320]), and the trained model’s remaining excess is +0.725 ([+0.597, +0.836]), both over 35 families with 10,000 resamples. Predictions and criteria were fixed before either ablated arm was computed.

Across the other nine models, one holds a buffer encoding anything about base pairing. The rest carry position indices, rotary frequencies, or all-zero token-type vectors, which survive randomization and encode where a nucleotide sits rather than what it pairs with. Two randomly initialized controls nonetheless sit above their own chance rates by 0.095 (RiNALMo) and 0.084 (RNA-FM), and no buffer in either accounts for it.

2.5 Multi-sequence replication

Six models show $CV < 6\%$: UTR-LM (0.025), SpliceBERT (0.026), RiNALMo (0.037), NT v2 (0.046), ERNIE-RNA (0.057), and Caduceus (0.057). The stem–loop sensitivity ratio is stable across biological replicates for these models.

RNA-FM ($CV = 0.338$) and Evo (0.244) show high sequence-to-sequence variability. Both also fail composition control (Section 2.1), consistent with raw sensitivity driven by sequence-specific composition rather than a stable structural signal.

2.6 Robustness analyses

The dinucleotide-stratified null absorbs the sole family (tRNA-Ala) surviving the nucleotide-stratified null for RNA-FM; second-order sequence context accounts for the residual signal. Replacing mean aggregation with median changes PS estimates by $< 5\%$ for RiNALMo and ERNIE-RNA and does not alter the model ranking. Using the last layer instead of the best layer reduces RiNALMo PS by 12% and ERNIE-RNA PS by 8%; both remain above the derangement null in $> 90\%$ of gate-qualifying families. Restricting to Watson–Crick pairs only (excluding wobble G–U pairs) does not

Table 5: Multi-sequence replication: within-family coefficient of variation (CV) of the stem-loop sensitivity ratio across all valid Rfam seed-alignment sequences (51 families, 253 sequences, 3–5 per family).

Model	Mean ratio	Mean CV	Families scored
RNA-FM	1.892	0.338	51/51
Evo	1.478	0.244	51/51
ERNIE-RNA	1.253	0.057	51/51
Caduceus	1.203	0.057	51/51
HyenaDNA	1.170	0.092	51/51
RiNALMo	1.133	0.037	51/51
NT v2 ^a	1.092	0.046	50/51
UTR-LM	1.059	0.025	51/51
SpliceBERT	1.047	0.026	51/51
DNABERT-2	0.990	0.100	51/51

^aOne Corona_5UTR sequence excluded (numerical overflow in ratio denominator); remaining 50 families scored normally.

change the primary conclusions. Widening the PS neighbor window from ± 1 to ± 2 positions reduces PS magnitudes uniformly across models without altering their relative ranking. A transversion control (purine $\rightarrow$ pyrimidine mutations, which guarantee disrupted pairing) produces ratios within 1.2% of the Watson–Crick complement ratios for five of six models tested, confirming that reversed base pairs differ from correct pairs enough to register as embedding perturbations. Nine of ten randomized-weight controls produce $|\text{PS}| < 10^{-8}$; DNABERT-2’s randomized PS (-7.1×10^{-3}) is comparable to its trained value, consistent with a tokenization artifact.

Family-level results and full robustness tables appear in Supplementary Tables S2–S4.

Eight pre-registered hypotheses (SHA-256 hashed before data collection) produced five confirmations and three failures. The pre-registered prediction that RNA-pretrained models would outperform DNA-pretrained models on partner specificity failed ($r_b = -0.28$, $p = 0.265$): Caduceus (14M, DNA) exceeds three RNA models.

2.7 Covariation versus geometry

Partner specificity could reflect sequence covariation rather than learned geometric pairing. Compensatory mutations that maintain base pairing are overrepresented in evolutionary RNA alignments; a model trained on such sequences might predict that mutating one position perturbs its covarying partner without encoding the physical mechanism of pairing.

To test this, we generated synthetic sequences for each Rfam family: the dot-bracket structure is preserved exactly, but nucleotides are assigned randomly (uniform Watson–Crick pairs at paired positions, uniform random at unpaired). These families share structure with their natural counterparts but carry no evolutionary covariation. ERNIE-RNA’s mean PS drops from 0.120 (natural) to 0.035 (synthetic), a 71% reduction. RiNALMo drops from 0.215 to 0.061, a 71% reduction. Both models lose roughly 70% of their PS when covariation is removed. The residual synthetic PS—still orders of magnitude above the weak-signal models—places an upper bound on the non-covariation contribution, since the synthetic sequences also destroy sequence context and positional composition gradients present in natural RNA.

As an independent check, we computed mutual information (MI) between paired positions from

Rfam seed alignments (51 families, $n \geq 3$ sequences each). MI cleanly separates paired from unpaired positions (mean MI = 0.349 vs 0.063; Wilcoxon $p = 3.5 \times 10^{-10}$), confirming that evolutionary covariation is detectable in the alignments. Per-family mean MI does not correlate significantly with model PS (ERNIE-RNA: $\rho = 0.320$, $p = 0.065$; RiNALMo: $\rho = 0.265$, $p = 0.130$). Families with strong covariation signal do not systematically produce higher PS, suggesting that the models’ partner specificity reflects learned structure rather than recapitulated covariation.

2.8 HTT case study

The CAG repeat region in huntingtin exon 1 forms hairpin structures whose stability scales with repeat count [Mezer et al., 2011, Krzyżosiak et al., 2011]. We evaluated five HTT exon 1 fragments (NM.002111.7) with CAG repeat counts of 17, 21, 36, 40, and 60; structures were predicted with ViennaRNA RNAfold [Lorenz et al., 2011]. Four models achieve $\geq 97\%$ probing accuracy on these predicted structures (Table 6), but the repeat region contains only C, A, and G—no U—creating extreme composition bias that linear probes exploit.

Table 6: HTT CAG-repeat case study. High probing accuracy coexists with null composition-controlled signal. Attn ρ : Spearman correlation between self-attention weights and the base-pairing contact map. Probing: linear probe accuracy for stem-vs-loop classification. Structures are RNAfold predictions, not experimentally determined.

Model	Mean ratio	> null	Attn ρ	Probing	WT→CAG60
RNA-FM	1.794	2/5	0.036	0.813	0.000
RiNALMo	1.031	0/5	0.054	0.985	0.068
NT v2	— [†]	—	0.140	0.636	0.013
ERNIE-RNA	1.086	0/5	0.070	0.985	0.023
SpliceBERT	0.996	0/5	0.036	0.977	0.042
UTR-LM	1.151	1/5	0.039	0.974	0.016

[†]6-mer tokenization yields degenerate embeddings on repetitive CAG. WT→CAG60 = cosine distance from CAG17 reference.

RNA-FM produces embedding distances from wild-type below 10^{-4} for all repeat counts, failing to distinguish normal from pathogenic alleles. RiNALMo shows the strongest separation (0.068 at CAG60), with distances increasing monotonically with repeat count. High probing accuracy on a computationally predicted structure can coexist with null composition-controlled signal: apparent structure decoding persists in a strongly composition-biased context.

3 Discussion

Composition-matched nulls absorb most of the apparent stem-loop sensitivity across all ten models. Two RNA-pretrained models—RiNALMo and ERNIE-RNA—retain reproducible partner-specific responses under the most stringent test. Multi-sequence replication shows low within-family variance across eight of ten models.

The underlying problem is general: whenever a structural annotation correlates with feature composition, any embedding metric sensitive to feature identity inherits the correlation. The nucleotide-stratified permutation null controls for this by breaking the structure–composition link while preserving nucleotide identity. The same template applies to any biological structure correlated with sequence composition.

RiNALMo and ERNIE-RNA reach partner specificity through different pathways. ERNIE-RNA injects a base-pairing attention bias; its randomized control shows this bias provides capacity without producing signal on its own. RiNALMo uses standard attention without structural bias and achieves higher PS at $7.5\times$ the parameters. Nine of ten randomized-weight controls produce $|\text{PS}| < 10^{-8}$, supporting a learned origin; DNABERT-2’s non-negligible randomized PS (-7.1×10^{-3}) confirms that its trained PS reflects tokenization rather than structure. Whether this partner specificity reflects a geometric internal model of RNA folding or a simpler statistical regularity remains open. A synthetic covariation control (Section 2.7) shows that roughly 70% of PS reflects evolutionary signal; the residual $\sim 30\%$ provides an upper bound on learned geometric sensitivity.

Neither training domain nor architecture family alone predicts composition-controlled sensitivity. The three SSM-based models (Evo, Caduceus, HyenaDNA) span nearly the full range of mean ratios (1.17–1.48), as do the seven transformers (0.99–1.89). RNA-pretrained models occupy four of the top five positions, but Evo and Caduceus—both DNA-pretrained—outperform three of the five RNA models. The two models that survive the partner specificity test (RiNALMo and ERNIE-RNA) share RNA pretraining but differ in mechanism: ERNIE-RNA injects explicit base-pairing attention bias, while RiNALMo uses standard self-attention at $7.5\times$ the parameters. Scale alone does not explain the result: Evo at 7B parameters produces high raw ratios but high within-family variance, and its signal is largely absorbed by composition nulls. What distinguishes the two positive models from the remaining eight is an open question whose answer likely requires internal mechanistic analysis rather than further behavioral benchmarking.

RNA structure prediction benchmarks have moved toward stricter evaluation because high apparent performance can fail to generalize across families [Szikszai et al., 2022]. This work extends the concern from dataset similarity to sequence-composition confounding. NT v2 and DNABERT-2 show near-zero embedding-level partner specificity despite passing many families at the composition gate, suggesting that structure-relevant information in these models, if present, may reside in attention patterns that embedding-level metrics do not capture.

Limitations. The complement-swap metric tests sensitivity to single base-pair disruptions; it does not test tertiary contacts, pseudoknots, or long-range interactions. The synthetic covariation control (Section 2.7) accounts for roughly 70% of PS, but the synthetic sequences also destroy sequence context and positional gradients, so the residual may overestimate true geometric sensitivity. HTT structures are RNAfold predictions. Ten models do not cover the full model space. Pretraining-set contamination cannot be fully reconstructed. Multi-nucleotide tokenization limits direct comparability for NT v2 and DNABERT-2.

Practical recommendations. Any claim that an RNA representation encodes secondary structure should report stem–loop sensitivity alongside a nucleotide-stratified null, so readers can separate composition sensitivity from structure awareness. Partner specificity is more informative than aggregate stem–loop contrasts: a model that perturbs stems more than loops may respond to GC content, while one that selectively perturbs the annotated pairing partner encodes positional pairing information. Randomized-weight controls should establish each architecture’s baseline capacity. Replication across multiple sequences per family guards against single-sequence artifacts; a coefficient of variation above 10% signals instability. For multi-nucleotide tokenizers, the mapping from token-level to position-level metrics should be stated explicitly.

4 Materials and Methods

4.1 Benchmark construction

We evaluate 52 Rfam 14.10 families [Kalvari et al., 2020] spanning ribozymes, riboswitches, snRNAs, cis-regulatory elements, miRNA precursors, CRISPR repeats, and other ncRNAs. Positions are labeled stem or loop from consensus dot-bracket annotation. Minimum inclusion criteria: ≥ 20 nt, ≥ 5 stem and ≥ 5 loop positions. Pseudoknots, noncanonical pairs, and gap positions are excluded.

All structures are consensus secondary structures from Rfam seed alignments; none are experimentally determined crystal or cryo-EM structures. Pretraining data overlap cannot be fully reconstructed for all models. The full family list with accessions appears in Supplementary Table S1.

4.2 Models and controls

Table 7: Models evaluated.

Model	Domain	Params	Architecture	Tokenization
RNA-FM [Chen et al., 2022]	RNA	99M	Transformer	Character
RiNALMo [Penić et al., 2025]	RNA	650M	Transformer	Character
ERNIE-RNA [Yin et al., 2024]	RNA	86M	Transformer [†]	Character
SpliceBERT [Chen et al., 2024]	RNA	19M	Transformer	Character
UTR-LM [Chu et al., 2024]	RNA	1.2M	Transformer	Character
NT v2 [Dalla-Torre et al., 2024]	DNA	56M	Transformer	6-mer
DNABERT-2 [Zhou et al., 2023]	DNA	117M	Transformer	BPE
Caduceus [Schiff et al., 2024]	DNA	14M	Mamba (SSM)	Character
HyenaDNA [Nguyen et al., 2023]	DNA	5.4M	Hyena (SSM)	Character
Evo [Nguyen et al., 2024]	DNA	7B	StripedHyena	Character

[†]ERNIE-RNA injects a base-pairing attention bias from the second layer onward.

We evaluate ten models covering attention, SSM, and hybrid architectures (Table 7): five pretrained on RNA, five on DNA. Randomized-weight controls for all ten models reinitialize all learned parameters (Xavier normal for matrices, $\mathcal{N}(0, 0.02)$ for vectors) while preserving architecture. NT v2 and DNABERT-2 use multi-nucleotide tokenization (6-mer and BPE), limiting direct comparability for position-level metrics.

4.3 Perturbation tests

For each annotated position, we swap the nucleotide with its Watson–Crick complement ($A \leftrightarrow U$, $C \leftrightarrow G$) and measure the cosine distance between wild-type and mutant embeddings at that position, maximized over layers. Cosine distance is scale-invariant across architectures. For multi-nucleotide tokenization, token-level outputs are mapped to nucleotide positions by distributing the token embedding equally across constituent nucleotides.

Nucleotide-stratified null. Stem/loop labels are shuffled within each nucleotide type (1,000 permutations), preserving nucleotide identity while breaking its structural label. Layer selection is repeated inside every permutation. A dinucleotide-stratified extension additionally conditions on the 3' neighbor.

Partner specificity. For each Watson–Crick pair (i, j) where j is the annotated partner:

$$\text{PS}(i, j) = \Delta_j - \max(\Delta_{j-1}, \Delta_{j+1})$$

where Δ_k is the cosine distance at position k after the complement swap at position i . Positive PS indicates the model perturbs the true partner more than its immediate neighbors (chance = $1/3$). Partner assignments are tested against a within-stem derangement null (500 derangements), controlling for positional and compositional confounds.

Covariation control. To distinguish learned geometric sensitivity from recapitulated evolutionary covariation, synthetic sequences were generated for each family: dot-bracket structure preserved exactly, nucleotides assigned randomly (uniform Watson–Crick pairs at paired positions, uniform random at unpaired). Mutual information (MI) between paired and unpaired positions was computed from Rfam seed alignments as an independent covariation measure.

4.4 Statistical analysis and replication

The biological unit of replication is the RNA family. The primary outcome is partner specificity (PS); the primary comparison is trained vs. randomized-weight PS. A family is *gate-qualifying* for a given model if its raw stem-loop sensitivity ratio exceeds the nucleotide-stratified null at the 95th percentile. PS is computed only over gate-qualifying families and aggregated as their mean.

To test stability across biological replicates, we repeat the full evaluation on all valid sequences from Rfam 14.10 seed alignments: 253 sequences across 51 families (3–5 per family). Within-family coefficient of variation (CV) quantifies replication stability.

Family-level bootstrap resampling (10,000 iterations, seed 42) provides 95% CIs for mean PS and mean sensitivity ratio. For the binary gate test, each family’s ratio is compared to its own 95th-percentile permutation null; we report the count exceeding this threshold alongside the expected false-positive count under independence ($m \times \alpha = 52 \times 0.05 = 2.6$). Bonferroni correction ($\alpha/m \approx 0.001$) provides a conservative bound.

Data and Code Availability

All data, scripts, and pre-registration protocols are available at <https://github.com/elliotttower/rna-structure-audit> and archived at Zenodo (DOI: 10.5281/zenodo.21362717).

The deposited panel holds one directory per model, each stamped with the commit and library versions it was computed under. Preregistrations are frozen at a commit hash, deviations are dated in `DEVIATIONS.md`, and reported numbers are bound to their runs, using `prereg`, `results` and `citations` [Tower, 2026].

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Ethics approval and consent to participate. This study uses exclusively publicly available pretrained model weights and published RNA sequences. No human subjects or animal subjects were involved. No ethical approval was required.

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